**DBMS Tables Normalization Lab**

**Question 1**

**Problem Statement:**

Normalize the table
up to 3NF.

**Dataset:**

| OrderID | CustomerName | Product | Price |
|---|---|---|---|
| 1 | Anuj | Laptop | 50000 |
| 1 | Anuj | Mouse | 500 |
| 2 | Amit | Laptop | 50000 |
| 2 | Amit | Keyboard | 1500 |
| 3 | Ravi | Mouse | 500 |
| 4 | Sita | Laptop | 50000 |
| 5 | Mohan | Keyboard | 1500 |
| 6 | Riya | Mouse | 500 |
| 7 | Karan | Laptop | 50000 |
| 8 | Neha | Keyboard | 1500 |

**Solution:**

**Orders**

| OrderID | CustomerName |
|---|---|
| 1 | Anuj |
| 2 | Amit |
| 3 | Ravi |
| 4 | Sita |
| 5 | Mohan |
| 6 | Riya |
| 7 | Karan |
| 8 | Neha |

**OrderDetails**

| OrderID | Product |
|---|---|
| 1 | Laptop |
| 1 | Mouse |
| 2 | Laptop |
| 2 | Keyboard |
| 3 | Mouse |
| 4 | Laptop |
| 5 | Keyboard |
| 6 | Mouse |

| | |
|---|---|
| 7 | Laptop |
| 8 | Keyboard |

**Product**

| Product | Price |
|---|---|
| Laptop | 50000 |
| Mouse | 500 |
| Keyboard | 1500 |

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**Question 2**

**Problem Statement:**

Identify dependencies and normalize to 3NF.

**Dataset:**

| EmpID | EmpName | DeptID | DeptName |
|---|---|---|---|
| 101 | Anuj | D1 | HR |
| 102 | Amit | D2 | IT |
| 103 | Ravi | D1 | HR |
| 104 | Sita | D3 | Finance |
| 105 | Mohan | D2 | IT |
| 106 | Riya | D1 | HR |
| 107 | Karan | D3 | Finance |
| 108 | Neha | D2 | IT |
| 109 | Pooja | D1 | HR |
| 110 | Rahul | D3 | Finance |

**Solution :**

**Employee**

| EmpID | EmpName | DeptID |
|---|---|---|
| 101 | Anuj | D1 |
| 102 | Amit | D2 |
| 103 | Ravi | D1 |
| 104 | Sita | D3 |
| 105 | Mohan | D2 |
| 106 | Riya | D1 |
| 107 | Karan | D3 |
| 108 | Neha | D2 |

| | | |
|---|---|---|
| 109 | Pooja | D1 |
| 110 | Rahul | D3 |

**Department :**

| DeptID | DeptName |
|---|---|
| D1 | HR |
| D2 | IT |
| D3 | Finance |

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**Question 3**

**Problem Statement:**

Remove transitive dependency and normalize.

**Dataset:**

| StudentID | Subject | Teacher | TeacherPhone |
|---|---|---|---|
| 1 | Math | Ravi | 9999 |
| 1 | Science | Ravi | 9999 |
| 2 | English | Sita | 8888 |
| 3 | Math | Ravi | 9999 |
| 4 | Science | Ravi | 9999 |
| 5 | English | Sita | 8888 |
| 6 | Math | Ravi | 9999 |
| 7 | Science | Ravi | 9999 |
| 8 | English | Sita | 8888 |
| 9 | Math | Ravi | 9999 |

**Solution :**

**Student :**

| StudentID | Subject |
|---|---|
| 1 | Math |
| 1 | Science |
| 2 | English |
| 3 | Math |
| 4 | Science |
| 5 | English |

| | |
|---|---|
| 6 | Math |
| 7 | Science |
| 8 | English |
| 9 | Math |

**TeacherSubject :**

| Subject | Teacher |
|---|---|
| Math | Ravi |
| Science | Ravi |
| English | Sita |

**TeacherPhone :**

| Teacher | TeacherPhone |
|---|---|
| Ravi | 9999 |
| Sita | 8888 |

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**Question 4**

**Problem Statement:**

Normalize to 3NF and identify all dependencies.

**Dataset:**

| BookID | Author | Publisher | PublisherAddress |
|---|---|---|---|
| B1 | A1 | P1 | Delhi |
| B2 | A2 | P2 | Mumbai |
| B3 | A1 | P1 | Delhi |
| B4 | A3 | P3 | Pune |
| B5 | A2 | P2 | Mumbai |
| B6 | A1 | P1 | Delhi |
| B7 | A3 | P3 | Pune |
| B8 | A2 | P2 | Mumbai |
| B9 | A1 | P1 | Delhi |
| B10 | A3 | P3 | Pune |

**Solution :**

**Book :**

| BookID | Author | Publisher |
|---|---|---|

| | | |
|---|---|---|
| B1 | A1 | P1 |
| B2 | A2 | P2 |
| B3 | A1 | P1 |
| B4 | A3 | P3 |
| B5 | A2 | P2 |
| B6 | A1 | P1 |
| B7 | A3 | P3 |
| B8 | A2 | P2 |
| B9 | A1 | P1 |
| B10 | A3 | P3 |

**Publisher :**

| Publisher | PublisherAddress |
|---|---|
| P1 | Delhi |
| P2 | Mumbai |
| P3 | Pune |

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**Question 5**

**Problem Statement:**

Find candidate key and normalize to 3NF.

**Dataset:**

| OrderID | ProductID | ProductName | CustomerID | CustomerName |
|---|---|---|---|---|
| 1 | P1 | Laptop | C1 | Anuj |
| 1 | P2 | Mouse | C1 | Anuj |
| 2 | P1 | Laptop | C2 | Amit |
| 2 | P3 | Keyboard | C2 | Amit |
| 3 | P2 | Mouse | C3 | Ravi |
| 4 | P1 | Laptop | C4 | Sita |
| 5 | P3 | Keyboard | C5 | Mohan |
| 6 | P2 | Mouse | C6 | Riya |
| 7 | P1 | Laptop | C7 | Karan |
| 8 | P3 | Keyboard | C8 | Neha |

**Solution :**

**Order :**

| OrderID | CustomerID |
|---|---|

| | |
|---|---|
| 1 | C1 |
| 2 | C2 |
| 2 | C2 |
| 3 | C3 |
| 4 | C4 |
| 5 | C5 |
| 6 | C6 |
| 7 | C7 |
| 8 | C8 |

**Customer :**

| CustomerID | CustomerName |
|---|---|
| C1 | Anuj |
| C2 | Amit |
| C2 | Amit |
| C3 | Ravi |
| C4 | Sita |
| C5 | Mohan |
| C6 | Riya |
| C7 | Karan |
| C8 | Neha |

**Product :**

| ProductID | ProductName |
|---|---|
| P1 | Laptop |
| P2 | Mouse |
| P3 | Keyboard |

**OrderDetails :**

| OrderID | ProductID |
|---|---|
| 1 | P1 |
| 1 | P2 |
| 2 | P1 |
| 2 | P3 |
| 3 | P2 |

| | |
|---|---|
| 4 | P1 |
| 5 | P3 |
| 6 | P2 |
| 7 | P1 |
| 8 | P3 |

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**Question 6**

**Problem Statement:**

Normalize the student-course table up to 3NF.

**Dataset:**

| StudentID | StudentName | Course | Instructor |
|---|---|---|---|
| 1 | Anuj | Math | Ravi |
| 1 | Anuj | Science | Ravi |
| 2 | Amit | English | Sita |
| 3 | Ravi | Math | Ravi |
| 4 | Sita | Science | Ravi |
| 5 | Mohan | English | Sita |
| 6 | Riya | Math | Ravi |
| 7 | Karan | Science | Ravi |
| 8 | Neha | English | Sita |
| 9 | Pooja | Math | Ravi |

**Solution :**

**Student :**

| StudentID | StudentName |
|---|---|
| 1 | Anuj |
| 2 | Amit |
| 3 | Ravi |
| 4 | Sita |
| 5 | Mohan |
| 6 | Riya |
| 7 | Karan |
| 8 | Neha |
| 9 | Pooja |

**Course :**

| Course | Instructor |
|---|---|
| Math | Ravi |
| Science | Ravi |
| English | Sita |

**Enrollment :**

| StudentID | Course |
|---|---|
| 1 | Math |
| 1 | Science |
| 2 | English |
| 3 | Math |
| 4 | Science |
| 5 | English |
| 6 | Math |
| 7 | Science |
| 8 | English |
| 9 | Math |

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**Question 7**

**Problem Statement:**

Normalize employee-project data to 3NF.

**Dataset:**

| EmpID | ProjectID | ProjectName | Manager |
|---|---|---|---|
| 101 | P1 | Alpha | Raj |
| 101 | P2 | Beta | Raj |
| 102 | P1 | Alpha | Raj |
| 103 | P3 | Gamma | Simran |
| 104 | P2 | Beta | Raj |
| 105 | P3 | Gamma | Simran |
| 106 | P1 | Alpha | Raj |
| 107 | P2 | Beta | Raj |
| 108 | P3 | Gamma | Simran |
| 109 | P1 | Alpha | Raj |

**Solution :**

**EmployeeProject :**

| EmpID | ProjectID |
|---|---|
| 101 | P1 |

| | |
|---|---|
| 101 | P2 |
| 102 | P1 |
| 103 | P3 |
| 104 | P2 |
| 105 | P3 |
| 106 | P1 |
| 107 | P2 |
| 108 | P3 |
| 109 | P1 |

**Project :**

| ProjectID | ProjectName | Manager |
|---|---|---|
| P1 | Alpha | Raj |
| P2 | Beta | Raj |
| P3 | Gamma | Simran |

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**Question 8**

**Problem Statement:**

Normalize sales data up to 3NF.

**Dataset:**

| SaleID | Product | Category | Price |
|---|---|---|---|
| 1 | Laptop | Electronics | 50000 |
| 2 | Mouse | Electronics | 500 |
| 3 | Keyboard | Electronics | 1500 |
| 4 | Shirt | Clothing | 1000 |
| 5 | Jeans | Clothing | 2000 |
| 6 | Laptop | Electronics | 50000 |
| 7 | Mouse | Electronics | 500 |
| 8 | Shirt | Clothing | 1000 |
| 9 | Jeans | Clothing | 2000 |
| 10 | Keyboard | Electronics | 1500 |

**Solution :**

**Product :**

| Product | Category | Price |
|---|---|---|
| Laptop | Electronics | 50000 |

| | | |
|---|---|---|
| Mouse | Electronics | 500 |
| Keyboard | Electronics | 1500 |
| Shirt | Clothing | 1000 |
| Jeans | Clothing | 2000 |

**Sales :**

| SaleID | Product |
|---|---|
| 1 | Laptop |
| 2 | Mouse |
| 3 | Keyboard |
| 4 | Shirt |
| 5 | Jeans |
| 6 | Laptop |
| 7 | Mouse |
| 8 | Shirt |
| 9 | Jeans |
| 10 | Keyboard |

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**Question 9**

**Problem Statement:**

Normalize hospital patient records to 3NF.

**Dataset:**

| PatientID | PatientName | Doctor | DoctorSpecialization |
|---|---|---|---|
| 1 | Anuj | Dr. Ravi | Cardiology |
| 2 | Amit | Dr. Sita | Neurology |
| 3 | Ravi | Dr. Ravi | Cardiology |
| 4 | Sita | Dr. Sita | Neurology |
| 5 | Mohan | Dr. Ravi | Cardiology |
| 6 | Riya | Dr. Sita | Neurology |
| 7 | Karan | Dr. Ravi | Cardiology |

| | | | |
|---|---|---|---|
| 8 | Neha | Dr. Sita | Neurology |
| 9 | Pooja | Dr. Ravi | Cardiology |
| 10 | Rahul | Dr. Sita | Neurology |

**Solution :**

**Patient :**

| PatientID | PatientName | Doctor |
|---|---|---|
| 1 | Anuj | Dr. Ravi |
| 2 | Amit | Dr. Sita |
| 3 | Ravi | Dr. Ravi |
| 4 | Sita | Dr. Sita |
| 5 | Mohan | Dr. Ravi |
| 6 | Riya | Dr. Sita |
| 7 | Karan | Dr. Ravi |
| 8 | Neha | Dr. Sita |
| 9 | Pooja | Dr. Ravi |
| 10 | Rahul | Dr. Sita |

**Doctor :**

| Doctor | DoctorSpecialization |
|---|---|
| Dr. Ravi | Cardiology |
| Dr. Sita | Neurology |

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**Question 10**

**Problem Statement:**

Normalize university data to 3NF.

**Dataset:**

| StudentID | Dept | HOD |
|---|---|---|
| 1 | CSE | Dr. A |
| 2 | IT | Dr. B |
| 3 | CSE | Dr. A |
| 4 | ECE | Dr. C |
| 5 | IT | Dr. B |
| 6 | CSE | Dr. A |

| | | |
|---|---|---|
| 7 | ECE | Dr. C |
| 8 | IT | Dr. B |
| 9 | CSE | Dr. A |
| 10 | ECE | Dr. C |

**Solution :**

**Student :**

| StudentID | Dept |
|---|---|
| 1 | CSE |
| 2 | IT |
| 3 | CSE |
| 4 | ECE |
| 5 | IT |
| 6 | CSE |
| 7 | ECE |
| 8 | IT |
| 9 | CSE |
| 10 | ECE |

**Department :**

| Dept | HOD |
|---|---|
| CSE | Dr. A |
| IT | Dr. B |
| ECE | Dr. C |